Paper III – Subspeciality of Urology (MCh Urology, Summer 2018)

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III-1. Embryology and management of ectopic ureter.

Type: LAQ (25 marks)

Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

Bud position on wolffian duct	Absorption	Orifice	Result
Too caudal	Early	High and lateral, short tunnel	VUR (vesicoureteral reflux)
Too cranial	Late or never	Low and medial (bladder neck, urethra) or stays on wolffian duct	**Ectopic ureter**, often obstructed

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

Sites of ectopia (the key sex difference)

	Female	Male
Sites (commonest first)	Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**)	Posterior urethra, seminal vesicle; also vas, ejaculatory duct

	Female	Male
Relation to sphincter	Can open **below** the external sphincter	Always **above** the external sphincter
Continence	**Continuous dribbling with normal voiding**	**Never incontinent**
Typical presentation	Constant dampness, perineal pus, UTI (urinary tract infection)	**Epididymo-orchitis**, UTI, flank pain, hydronephrosis

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

Clinical features

- **Prenatal hydronephrosis**: most are detected this way.
- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.
- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

Evaluation

Test	What it shows
Ultrasound (first)	Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele
VCUG (voiding cystourethrogram)	Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter
DMSA (dimercaptosuccinic acid)	Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed
MR (magnetic resonance) urography	Non-dilated upper pole, small dysplastic kidney, complex anatomy
Examination / cystovaginoscopy	Perineal orifice between urethra and vagina, cannulated for contrast

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

Management

Goals: preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

```
Neonate with sepsis or massive dilation
   → end cutaneous ureterostomy (or nephrostomy); DMSA later
DUPLEX system
   Upper pole non-functioning → upper-pole heminephroureterectomy
   Upper pole worth keeping   → ureteroureterostomy (UU) / pyeloureterostomy
   + lower-pole VUR           → common sheath reimplant
                                (or lower-pole reimplant + distal UU)
SINGLE system
   Functioning     → ureteral reimplant (± tapering)
   Non-functioning → nephroureterectomy (laparoscopic)
   Bilateral       → bladder neck reconstruction ± augmentation
```

- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat. Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.
- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

Diagram to draw: duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

> **Key points to remember:**
> - Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
> - Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
> - DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
> - Protect the lower-pole blood supply; stent the recipient ureter for UU.

III-2. Discuss the role of tissue typing and cross match in live and cadaveric

Type: LAQ (25 marks)

The question is incomplete; it is taken to mean "... live and cadaveric (deceased-donor) renal transplantation".

Introduction

- Two immune barriers decide compatibility: **preformed antibodies** (anti-A/B isoagglutinins and anti-HLA (human leukocyte antigen) antibodies) and **T-cell responses** to mismatched HLA.
- **Tissue typing** measures how well donor and recipient HLA match. **Antibody screening and crossmatch** detect preformed antibodies before the kidney meets them. Together they prevent **hyperacute rejection**, now practically eliminated.

The HLA system

- HLA is the human MHC (major histocompatibility complex), a gene cluster on **chromosome 6 (short arm)**. Genes are inherited as two **haplotypes** (one from each parent) and expressed codominantly.

	Class I	Class II
Loci	HLA-A, -B, -C	HLA-DR, -DQ, -DP
Expression	All nucleated cells	Antigen-presenting cells, endothelium
Presents to	CD8 (cytotoxic) T cells	CD4 (helper) T cells

- **Sibling** chance: 25% two-haplotype identical, 50% one-haplotype, 25% none. A parent or child always shares one haplotype.

Tissue typing methods

Method	Principle	Note
Serological (microlymphocytotoxicity)	Lymphocytes + typed antisera + complement; dead cells take up dye	Low resolution; largely replaced
Molecular (DNA-based)	PCR (polymerase chain reaction) with SSP (sequence-specific primers), PCR-SSO (sequence-specific oligonucleotide probes), sequence-based typing, **NGS** (next-generation sequencing)	High (allele-level) resolution; now standard

- **Matching:** usually reported as mismatches at **A, B and DR** (0–6). **DR matters most**, then B, then A; DQ increasingly typed. Fewer mismatches mean better graft survival and less sensitisation for a future transplant (Collaborative Transplant Study, Opelz). The effect is larger in deceased-donor grafts.

ABO compatibility

- Same rules as transfusion: **O universal donor, AB universal recipient**.
- **A2 (non-A1) donors** can go to O or B recipients with low anti-A titre.
- **ABO-incompatible** living transplant is possible: lower titres (plasmapheresis or immunoadsorption, rituximab, IVIG (intravenous immunoglobulin)) to a centre target before surgery.

Sensitisation and antibody screening

- Anti-HLA antibodies come from **pregnancy, blood transfusion, previous transplant** (and infection).
- **PRA (panel reactive antibody):** recipient serum tested against a panel of lymphocytes; result as % of panel killed.
- **Solid-phase (Luminex) single-antigen beads:** identify each antibody and its strength (MFI, mean fluorescence intensity). Antibodies against a donor are **DSA (donor-specific antibodies)**.
- **cPRA (calculated PRA):** % of donors the patient would be incompatible with, calculated from listed "unacceptable antigens". It replaced cell-panel PRA for allocation; very high cPRA gets allocation priority.

Crossmatch

Type	Method	Meaning if positive
CDC (complement-dependent cytotoxicity)	Donor T and B lymphocytes + recipient serum + complement; cell death scored. **DTT** (dithiothreitol) removes IgM autoantibody; AHG (antihuman globulin) increases sensitivity	**Positive T-cell CDC with IgG DSA = absolute contraindication** (hyperacute rejection; Patel and Terasaki, 1969)
Flow cytometry crossmatch	Antibody binding on donor cells detected by fluorescent anti-IgG	More sensitive; CDC-negative/flow-positive = higher risk of AMR (antibody-mediated rejection); desensitise or exchange
Virtual crossmatch	Compare recipient's bead-defined antibodies with donor HLA type	Speeds deceased-donor allocation; confirm by cell-based crossmatch

- **B-cell positive** alone: class II antibody or non-specific binding; interpret with bead results.
- Crossmatch uses a **recent serum** plus historical peak sera.

Living versus deceased donor

	Living donor	Deceased (cadaveric) donor
Typing	ABO and HLA early in donor work-up; choose best of several donors	ABO and HLA at donor referral
Crossmatch	At work-up and a **final crossmatch shortly before surgery**	At the time of offer, often virtual first; done against the list rapidly (cold ischaemia)
Role of matching	Less critical: even unrelated living donors do better than well-matched deceased donors (Terasaki, 1995)	Allocation uses HLA match, sensitisation (cPRA), waiting time and donor quality (KDPI, Kidney Donor Profile Index)
If incompatible	**Paired (swap) exchange**, desensitisation, ABO-incompatible protocol	Patient waits; acceptable-mismatch programmes

- **India:** swap (paired) donation is legal under the Transplantation of Human Organs (THOA) amendment, 2011.

Managing the incompatible or sensitised recipient

- **Kidney paired donation**, including chains with altruistic donors.
- **Desensitisation:** plasmapheresis + low-dose IVIG, or high-dose IVIG, ± rituximab; **imlifidase** (cleaves IgG) for highly sensitised deceased-donor recipients (conditional EU approval). Desensitised HLA-incompatible living transplant gave better survival than waiting on dialysis (Montgomery, 2011).

After transplant

- Monitor for **de novo DSA** (especially class II) with dd-cfDNA (donor-derived cell-free DNA) and biopsy; de novo DSA predicts graft loss.

Recent advances

- **Eplet (epitope) matching**, high-resolution NGS typing, virtual crossmatch, imlifidase.

Diagram to draw: CDC crossmatch: donor lymphocytes + recipient serum + complement, dead cells take up dye (positive).

> **Key points to remember:**
> - HLA on chromosome 6; match A, B, DR; DR matters most.
> - Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
> - Flow and bead assays are more sensitive; cPRA guides allocation.
> - Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

III-3a. Treatment algorithm of undescended testis.

Type: SAQ (10 marks)

Principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe for 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy after 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery improves testicular growth and lowers cancer risk.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); never changes the operation; delays surgery.
- **No hormonal therapy** (hCG (human chorionic gonadotrophin) or LHRH (luteinising hormone-releasing hormone)): descent in about 20% only; hCG may increase germ-cell apoptosis.
- **Bilateral non-palpable testes** (especially with hypospadias or small phallus): urgent **karyotype and hormones** to exclude **CAH (congenital adrenal hyperplasia)** and other DSD (differences of sex development).

Algorithm

```
Boy with empty scrotum
  ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones,
electrolytes)
  └─ Otherwise → re-examine at 6 months corrected age
        Still undescended → refer → examination under anaesthesia (EUA)
          ├─ Retractile → yearly review (risk of ascent)
          ├─ Palpable → inguinal or scrotal (Bianchi) orchidopexy
          └─ Non-palpable → diagnostic laparoscopy
                ├─ Testis seen → laparoscopic orchidopexy
                │       short vessels → Fowler–Stephens (1 or 2 stage) / Shehata
                ├─ Vessels + vas enter ring → inguinal exploration (excise nubbin)
                └─ Blind-ending vessels → stop (vanished testis)
```

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.

Key operative points

- **Inguinal orchidopexy:** free the cord, ligate the processus, retroperitoneal mobilisation ± **Prentiss manoeuvre**, subdartos pouch.
- **Fowler–Stephens:** divides testicular vessels; testis survives on vasal collaterals; for testes not within 2–4 cm of the ring. Primary orchidopexy, when possible, is more successful.
- **Postpubertal intra-abdominal testis with normal contralateral testis: orchidectomy** (poor function, cancer risk); offer prosthesis.

Follow-up

- Check position and size at 6 months; teach **self-examination** after puberty (cancer risk 2–5 times).

Diagram to draw: laparoscopic findings at the internal ring (testis, vessels entering ring, blind-ending vessels).

Key points to remember:

- Operate from 6 months, preferably by 12 months and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy.
- Bilateral non-palpable: exclude CAH/DSD first.

III-3b. Collin's knife.

Type: SAQ (10 marks)

Taken to mean the Collings knife, the knife-shaped cutting electrode of the resectoscope (the reference chapters spell it both "Collins" and "Collings").

Description

- A **hot (electrosurgical) knife electrode** fitted to the **resectoscope working element** in place of the loop.
- Thin **straight or angled (L- or hook-shaped) blade**; cuts a precise line with **cutting current** under irrigation and direct vision.
- **Monopolar** versions need **non-conducting irrigant** (glycine, sorbitol or mannitol); bipolar versions allow normal saline.
- Adult and **infant (9.5 Fr) resectoscope** sizes.
- **Principle:** it **incises rather than resects**; a deep, controlled cut releases a fibrous ring with little tissue loss.

Uses (from the chapters)

Use	Technique point
TUIP (transurethral incision of prostate)	Gland under 30 mL, no median lobe; incise at 5 and 7 o'clock from below the ureteric orifice to just proximal to the verumontanum, down to the capsule
Bladder neck contracture (after TURP (transurethral resection of prostate) or simple prostatectomy)	Incise until the scar springs open, to about 22 Fr; mitomycin C for recurrence
Ureterocele incision (older children)	Low transverse full-thickness incision near the bladder floor to reduce reflux
PUV (posterior urethral valve) ablation	Infant resectoscope only if urethral calibre is normal; incise at 5, 7 (± 12) o'clock; avoid hot-loop resection (stricture)
Bladder diverticulum neck	Incise the narrow neck so the diverticulum drains
PCNL (percutaneous nephrolithotomy) tract	Incise scarred fascia when rigid dilators fail

Alternatives

- **Cold knife (Sachse)** urethrotome: preferred for urethral stricture; hot knife unnecessary there.
- **Holmium or thulium laser**, Bugbee electrode, hook electrode, **Mohan's valvotome** (for valves).

Precautions and complications

- Stay away from the **ureteric orifices** and external sphincter; control depth (perforation, extravasation).
- TUR (transurethral resection) syndrome with monopolar irrigant if veins open.
- **Retrograde ejaculation** after bladder neck incision; stricture if the urethra is too small for the sheath.

Diagram to draw: resectoscope tip with L-shaped knife electrode, and the 5 and 7 o'clock TUIP incisions.

Key points to remember:

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

III-3c. Autosomal recessive polycystic kidney disease.

Type: SAQ (10 marks)

Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

Clinical features

Age	Features
Fetus	Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia**
Neonate	Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early**
Child	Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy
Older child	**Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal

- No association with renal tumours.

Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

ARPKD versus ADPKD

	ARPKD	ADPKD
Gene	PKHD1 (chromosome 6)	PKD1 (16), PKD2 (4)
Age	Perinatal	30–50 years
Cysts from	Collecting duct	Whole nephron
Ultrasound	Symmetric, homogeneous, echogenic	Large discrete cysts, asymmetric
Liver	Fibrosis (always)	Cysts
Other	None	Berry aneurysm, mitral prolapse

Diagram to draw: cut kidney with radial dilated collecting ducts.

> **Key points to remember:**
>
> - PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
> - Huge homogeneous echogenic kidneys with oligohydramnios.
> - Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

III-3d. Embryology and Management of P.U. valve patient.

Type: SAQ (10 marks)

Embryology

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into rectum and **urogenital sinus**; wolffian ducts migrate to the verumontanum, leaving small folds.

Young type	Anatomy	Origin
I (about 95%)	Leaflets from the verumontanum, fused anteriorly	Abnormal anterior insertion of the wolffian duct ends
II	Folds from verumontanum to bladder neck	Not real (secondary hypertrophy)
III	Ring diaphragm with central hole	**Persistent urogenital membrane**

- **COPUM (congenital obstructing posterior urethral membrane):** one membrane, torn by the first catheter into type I leaflets.
- Fetal urine from 10–14 weeks: obstruction causes **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.

Antenatal

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios. **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.
6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply.

- **VUR (vesicoureteral reflux)** (50–80%) is secondary; treat the bladder, not the ureter.

Prognosis

Nadir creatinine (first year)	ESRD (end-stage renal disease) risk
< 0.4 mg/dL	Nil
0.4–0.69	2%
0.7–0.99	27%
> 1.0	Up to 100%

- Lifetime ESRD risk 20–50%.

Long-term care

- **Valve bladder** (Mitchell): polyuria, poor sensation and residual urine keep the upper tracts dilated. Timed and double voiding, anticholinergic or α-blocker, **CIC (clean intermittent catheterisation)**, **overnight drainage**, Mitrofanoff, rarely augmentation.

- **CKD (chronic kidney disease) care; transplantation** after urodynamics and bladder optimisation; graft survival similar.

Diagram to draw: type I leaflets versus type III membrane, with keyhole VCUG.

Key points to remember:

- Type I (95%): wolffian duct insertion; type III: persistent urogenital membrane.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: overnight drainage and CIC.

III-3e. CSIC.

Type: SAQ (10 marks)

Taken to mean CISC – clean intermittent self-catheterisation.

Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

Catheters

Type	Note
Plain PVC (polyvinyl chloride), reusable after washing	Cheap; common in India
Hydrophilic-coated single-use	Less urethral trauma; longer time to first UTI (urinary tract infection)
Compact or closed-system	Convenience, outdoor use

Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.
- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.

- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

Diagram to draw: neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

III-3f. Controversies in management of varicocele.

Type: SAQ (10 marks)

Background

- Dilated pampiniform veins: about 15% of men, 35% with primary and up to 80% with secondary infertility; mostly left-sided (taller venous column to the left renal vein, valve failure).
- Harm: **heat**, oxidative stress, **sperm DNA fragmentation**; progressive atrophy, lower testosterone.
- **Grades:** subclinical (Doppler only), I (palpable with Valsalva), II (palpable), III (visible).

The controversies

Question	Evidence	Current position
Does repair improve fertility?	Madgar RCT (randomised controlled trial, 1995): pregnancy 44% versus 10% at 1 year; meta-analyses mixed	Repair **clinical** varicocele with **abnormal semen** in an infertile couple with normal female partner (AUA/ASRM (American Urological Association / American Society for Reproductive Medicine))
Subclinical varicocele?	No benefit for semen or pregnancy	**Do not treat**
Normal semen?	No proven benefit	Not for fertility
Adolescent varicocele	Testis volume is a surrogate; link to adult fertility unproven	Observe; repair for **testis asymmetry above 20%**, pain, or abnormal semen (Tanner 5 or 18 years)
NOA (non-obstructive azoospermia)	Small series: sperm returns to ejaculate in some	No definitive evidence; may be considered before micro-TESE (microsurgical testicular sperm extraction)
Before IVF/ICSI (in vitro fertilisation / intracytoplasmic sperm injection)	Repair may improve live-birth rate or allow less invasive ART (assisted reproduction)	Discuss, especially with oligozoospermia
High DNA fragmentation	Repair lowers fragmentation	Reasonable indication
Hypogonadism	Repair raises testosterone in hypogonadal men	Emerging indication
Pain	Dull ache on standing; relief in most after microsurgery	Repair after other causes excluded

Choice of technique

Technique	Recurrence	Hydrocele	Comment
Microsurgical subinguinal/inguinal	Lowest (about 1%)	Near 0	**Best**: artery and lymphatics preserved
Retroperitoneal (Palomo)	High with artery spared	High with mass ligation	Children (artery ligation acceptable)
Laparoscopic	3–15%	Up to 12%	Bilateral; serious morbidity possible
Embolisation	4–25%	Nil	Outpatient; technical failure

- **Also know:** isolated right or non-decompressing varicocele → scan for renal tumour; recurrence diagnosed by upright colour Doppler.

Diagram to draw: cord cross-section at microsurgery: artery, lymphatics and vas preserved, veins ligated.

Key points to remember:

- Treat only palpable varicocele with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence and hydrocele.

Rapid Revision Sheet – Paper III

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

III-1. Embryology and management of ectopic ureter. (LAQ)

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

III-2. Discuss the role of tissue typing and cross match in live and cadaveric (LAQ)

- HLA on chromosome 6; match A, B, DR; DR matters most.
- Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
- Flow and bead assays are more sensitive; cPRA guides allocation.
- Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

III-3a. Treatment algorithm of undescended testis. (SAQ)

- Operate from 6 months, preferably by 12 months and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy.
- Bilateral non-palpable: exclude CAH/DSD first.

III-3b. Collin's knife. (SAQ)

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

III-3c. Autosomal recessive polycystic kidney disease. (SAQ)

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

III-3d. Embryology and Management of P.U. valve patient. (SAQ)

- Type I (95%): wolffian duct insertion; type III: persistent urogenital membrane.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: overnight drainage and CIC.

III-3e. CSIC. (SAQ)

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

III-3f. Controversies in management of varicocele. (SAQ)

- Treat only palpable varicocele with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence and hydrocele.